

1. (a) What is a sequence?

(b) What does it mean to say that $\lim_{n \rightarrow \infty} a_n = 3$?

(c) What does it mean to say that $\lim_{n \rightarrow \infty} a_n = \infty$?

2. What are the following limits? Illustrate with a graph.

(a) $\lim_{n \rightarrow \infty} a_n$, where $a_n = \cos(n\pi)$

(b) $\lim_{n \rightarrow \infty} a_n$, where $a_n = \cos((n + 1/2)\pi)$

3. List the first five terms of the sequence, $a_n = \frac{1}{3^{n+1}}$, $n = 1, 2, \dots$

4. Find a formula for the general term a_n of the sequence, assuming the pattern for the first few terms continues

(a) $\{2, 4, 6, 8, \dots\}$

(b) $\{1, 3, 5, 7, \dots\}$

(c) $\{1, -1, 1, -1, \dots\}$

(d) $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \dots\}$

(e) $\{4, -1, \frac{1}{4}, -\frac{1}{16}, \frac{1}{64}, -\frac{1}{256}, \dots\}$

6. Determine the limit of the following sequences or explain why it does not exist. Show all necessary work. Use the fact that if $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} f(n) = L$ (where n is restricted to integers).

(a) $a_n = \frac{1}{2^n}$

(b) $a_n = \frac{n-2}{n^2}$

(c) $a_n = \frac{\sin(n)}{n}$

(d) $a_n = \frac{n}{n+1}$

(e) $a_n = (-1)^n \frac{n}{n+1}$

(f) $a_n = n^{1/n}$

7. Give an example showing that if $\lim_{n \rightarrow \infty} f(n) = L$ (where n is restricted to integers), it is not necessarily true that $\lim_{x \rightarrow \infty} f(x) = L$ (where x is a continuous real variable).